

Luiz Henrique Paludo

W218052923

ITAI 1378

Professor Patricia McManus

June 09 2026

Image Processing Fundamentals

For me was amazing how computer see images as matrix of numbers, it is fascinating that my daughter picture that I used as an example on my notebook, the computer see only a chart of red colors and numbers.

Mathematical operations change the pixel values in the image. When operating these changes we create visual effects such as making the image brighter, sharper or shows more details.

For me the most difficult technique to understand was the Kernel because it use neighbor's pixels instead of changing one pixel at a time.

In today labs, I understand how images can be modified using math operations, Nano banana do similar tasks, but use AI to learn and apply the changes automatically instead of the manual way we did to apply filters and transformation.

Today's labs I think the perfect real-world application would be photo filters on instagram, enhance the quality of an image in programs like lightroom.

Not in a future project, but I use this all the time, I use the traditional way to enhance the quality and when I can reach my perfectionist goal, I ask AI tools to help me on improve that extra mile.

What aspect interest me more is the facial recognition, I would like to know more on how the computers can identify a person using image processing and AI.

This lab helped me to understand that image are made by pixels and numbers, and also showed me that I can enhance the quality of a photo just changing that parameters.

The question that I have is simply, How can a AI agent can get someone pictures and simple change his face with another human face?